

Smart Campus Network Deployment and Testing

Your team is part of the Network Operations Unit hired to design and implement the network for a small university campus.

The campus contains the following blocks:

1. Administrative Block – Staff & Management
2. Computer Science Department – Labs & Classrooms
3. Student Hostel
4. (Bonus) Server Block – Provides Web/FTP/DNS services across the network

Requirements

The university requires:

- Separate VLANs for each block to improve security and management.
- Inter-VLAN Routing so that controlled communication between VLANs is possible.
- Multiple Routers must be deployed for scalability and redundancy.
- Dynamic Routing (OSPF or EIGRP) must be implemented between the routers for automatic route advertisement and fault tolerance.
- Port Security must be implemented on access switches to restrict unauthorized devices:
- Application Layer Services such as a simple web server or chat application to test end-to-end connectivity.
- (Bonus) Fault tolerance using EtherChannel for link redundancy between routers and access switches.

Network Equipment to be Used

- Routers: Minimum of 3
 - Each router performs inter-VLAN routing for the VLANs it serves.
 - Routers are interconnected directly through the access switches and run dynamic routing.
- Switches:
 - 2 Switches – one serving Administrative + CS Department VLANs, the other serving the Hostel VLAN.
 - Each switch handles VLAN trunking, EtherChannel (if configured), and Port Security on all access ports.
- End Devices:
 - Minimum of 3 PCs per VLAN
 - 1 Server providing Web/Chat/DNS services.

IP Addressing

- Use private IP addressing and VLSM.
- The second octet of selected IP must include the last digits of the group leader's registration number.
 - Example format: 10.<REG_LAST>.<VLAN_ID>.0/24
 - Gateway for each VLAN: 10.<REG_LAST>.<VLAN_ID>.1

Deliverables

- A complete VLAN plan, IP addressing scheme, and Port Security configuration table (showing allowed MAC addresses or limits).
- Router configurations showing inter-VLAN routing and OSPF/EIGRP dynamic routing.
- Switch configurations with VLANs, trunking, EtherChannel (if used), and Port Security commands.
- A working Application Layer service (Web/Chat/DNS) to test connectivity between VLANs.
- Testing and verification steps (ping, traceroute, route tables) demonstrating full campus connectivity, dynamic routing, and Port Security operation (e.g., violation detection).